

①

min grey level = 0

max " " = 7

3	3	3	3	3	3	3	3
3	4	4	4	4	4	4	3
3	4	2	2	2	2	4	3
3	4	2	5	5	2	4	3
3	4	2	5	5	2	4	3
3	4	2	2	2	2	4	3
3	4	4	4	4	4	4	3
3	3	3	3	3	3	3	3

$$S_{ij} = \frac{r_{ij} - r_{min}}{r_{max} - r_{min}} \times (S_{max} - S_{min}) + S_{min}$$

$$= \frac{3-2}{5-2} \times (7-0) + 0$$

$$\boxed{r_0 = 2.33 = 2}$$

$$r_1 = \frac{4-2}{5-2} \times 7-0 + 0$$

$$\boxed{r_1 = 5}$$

$$\boxed{r_2 = 0}$$

$$\boxed{r_3 = 7}$$

$$r_{min} = 2, r_{max} = 5, S_{max} = 7, S_{min} = 0$$

2	2	2	2	2	2	2	2
2	5	5	5	5	5	5	2
2	5	0	0	0	0	5	2
2	5	0	7	7	0	5	2
2	5	0	7	7	0	5	2
2	5	0	0	0	0	5	2
2	5	5	5	5	5	5	2
2	2	2	2	2	2	2	2

2.	Pixel value	N_k	PDF	CDF	$7 \times \text{CDF}$	output gray level
	0	123	0.03	0.03	0.21	0
	1	78	0.02	0.05	0.35	0
	2	281	0.06	0.11	0.77	1
	3	417	0.10	0.21	1.47	1
	4	639	0.15	0.36	2.52	3
	5	1054	0.25	0.61	4.27	4
	6	816	0.199	0.8	5.6	6
	7	688	0.16	0.96	6.72	7

($N = 4096$)

Pixel value	0	1	3	4	6	7
N_k	201	698	639	1054	816	688

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31	30	29	28
16	26	25	24
24	25	26	16
28	29	30	31

4x4 5 bit image.

in binary form:

11111	11110	11101	11100
10000	11010	11001	11000
11000	11001	11010	10000
11100	11101	11110	11111

$$B_2 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Thresholding is set at 27

27 $\uparrow$ 1
 $\downarrow$ 0

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

④

50	55	70
60	65	75
80	90	95

after γ -correction
central pixel = ?

formula $S = C \cdot r^\gamma$

C = constant = 1
 $\gamma = 1.25$

$2^7 = 128$

$L-1 = 127$

output image
apply direct formula:

132	148	202
166	184	220
239	277	284

There are many
pixel which are out
of range (0-127)

$\times 255$

196	126	279
235	255	299
314	353	373

γ	130	147	198
	163	181	226
	235	272	291

$\times 255$	33	37	50
	42	46	55
	60	69	74

output

⑤

$S = \{ 15, 5 < r \leq 10 \}$
 r otherwise

Gray level slicing

2	3	5	5	6
5	3	9	11	12
12	11	1	8	1
7	7	9	0	1

$\rightarrow$

2	3	5	5	15
5	3	15	11	12
12	11	1	15	1
15	15	15	0	1

(input image)

(output image)

⑥

3-bit image

1	2	3
4	5	0
7	2	1

convert binary

001	010	011
100	101	000
111	010	001

LSB discarded

000	010	010
100	100	000
100	010	000

0	2	2
4	4	0
6	2	0

(output)

⑦ Using median filter, determine denoised the central pixel:-

120	126	161
130	10	141
60	70	238

increasing order
 $\rightarrow$ 10 60 70 120 126 130 141 161 238

120	126	161
130	126	141
60	70	238

output image

r_k	n_k	PDF	CDF	$r_k \times CDF$	S_k	r_k	n_k				$G(z_k)$
0	8	0.125	0.125	0.87	1	0	0	0	0	0	0
1	10	0.15	0.275	1.42	2	1	0	0	0	0	0
2	10	0.15	0.425	2.47	3	2	0	0	0	0	0
3	2	0.03	0.45	3.15	3	3	0	0	0	0	0
4	12	0.18	0.63	4.41	4	4	20	0.312	0.312	2.18	2
5	16	0.25	0.88	6.16	6	5	20	0.312	0.624	4.368	4
6	4	0.06	0.94	6.88	7	6	16	0.25	0.874	6.118	6
7	2	0.03	0.97	6.79	7	7	8	0.125	0.997	6.956	7
$N = 64$							64				

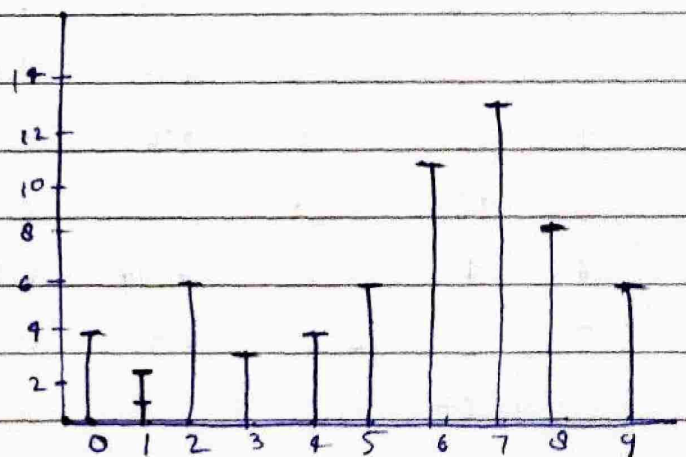
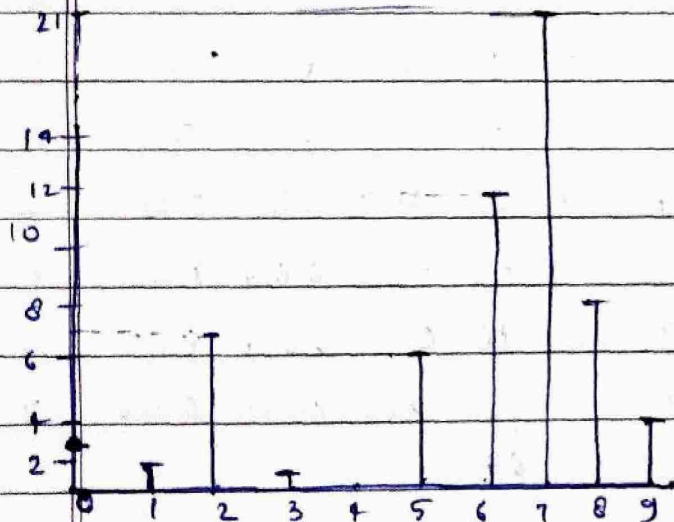
r_k	S_k	$G(z_k)$	z_k		
0	1	0	0	0	$\rightarrow$ 4
1	2	0	1	1	$\rightarrow$ 4
2	3	0	2	2	$\rightarrow$ 5
3	3	0	3	3	$\rightarrow$ 5
4	4	2	4	4	$\rightarrow$ 5
5	6	4	5	5	$\rightarrow$ 6
6	7	6	6	6	$\rightarrow$ 7
7	7	7	7	7	$\rightarrow$ 7

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r_k	n_k	PDF	CDF	$7 \times \text{CDF}$	
0	3	.046	.046	.322	0
1	2	.031	.077	.539	1
2	7	.109	.186	1.302	1
3	1	.015	.201	1.407	1
4	0	0	.201	1.407 2.058	1
5	6	.093	.294	3.367	2
6	12	.187	.481	5.613	3
7	21	.328	.809	6.538	6
8	8	.125	.934	6.538	7
9	4	.062	.996	6.972	7

Z_k	n_k				C_{r_k}
0	4	.062	.062	0.434	0
1	3	.046	.108	.706	1
2	6	.093	.201	1.407	1
3	3	.046	.247	1.729	2
4	4	.062	.309	2.163	2
5	6	.093	.402	2.814	3
6	11	.171	.573	4.011	4
7	13	.203	.776	5.4	5
8	8	.125	.901	6.3	6
9	6	.093	.994	6.9	7

64



r_k	s_k	C_{r_k}	Z_k
0	0	0	0
1	1	1	1
2	1	1	2
3	1	2	3
4	1	2	4
5	2	3	5
6	3	4	6
7	6	5	7
8	7	6	8
9	7	7	9

0	0
1	1
2	2
3	2
4	2
5	3
6	5
7	8
8	9
9	9

10) apply 3x3 median filter on shaded area.

20	30	50	80	100
30	20	80	100	110
25	225	70	0	120
30	30	80	100	130
40	50	90	125	140

sorted order for 225

(i) 20, ~~25~~, 30, 30, 80, 70, 80, 80, 225

(ii) for 70

0, 20, 30, 70, 80, 80, 100, 100, 225

(ii) 0

0, 70, 80, 80, 100, 100, 110, 120, 130

Final image:

20	30	50	80	100
30	20	80	100	110
25	30	80	100	120
30	30	80	100	130
40	50	90	125	140

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1st order derivative

- Response must be 0 in the areas of constant gray level
- Non-zero at the on-set of gray-level set of ramp.
- Non-zero along ramps.

2nd order derivative

- Zero in flat areas.
- Non-zero at on set and end-off gray level.
- Zero along ramps on constant slope.

12

50	50	50	50	50
50	50	50	50	50
10	10	10	10	10
10	10	10	10	10
10	10	10	10	10

L-4

Laplacian Mask:

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

for 0 value:

$$\begin{bmatrix} -100 & -50 & -50 & -50 & -100 \\ -90 & -40 & -40 & -40 & -90 \\ 30 & 40 & 40 & 40 & 30 \\ -10 & 0 & 0 & 0 & -10 \\ -20 & -10 & -10 & -10 & -20 \end{bmatrix}$$

for 1-value extension.

$$\begin{bmatrix} -98 & -99 & -99 & -99 & -98 \\ -89 & -90 & -90 & -90 & -89 \\ 31 & 40 & 40 & 40 & 31 \\ -9 & 0 & 0 & 0 & -9 \\ -18 & -9 & -9 & -9 & -18 \end{bmatrix}$$

for border value extension:

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ -90 & -90 & -90 & -90 & -90 \\ 40 & 40 & 40 & 40 & 40 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

output

(13)

6	6	6	6	5	4	3	2	1	1	1	1	1	1	6	6	6	6
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first order
derivative

0	0	0	-1	-1	-1	-1	-1	0	0	0	0	0	5	0	0	0	0
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2nd order
derivative

0	0	-1	0	0	0	0	+1	0	0	0	0	5	-5	0	0	0	0
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(14)

apply convolution on shaded pixel

3x3 mask

$$\begin{bmatrix} 0 & 1/6 & 0 \\ 1/6 & 1/3 & 1/6 \\ 0 & 1/6 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 30 & 40 & 50 & 70 & 90 \\ 40 & 50 & 80 & 60 & 100 \\ 35 & 255 & 70 & 0 & 120 \\ 30 & 45 & 80 & 100 & 130 \\ 40 & 50 & 90 & 125 & 140 \end{bmatrix}$$

⇒

30	40	50	70	90
40	86	65	62	100
85	118	93	59	120
30	84	78	89	130
40	50	90	125	140

output image